

PARTS LIST

Semiconductors:

D1	- BAT83 or 1N4148 signal diode
LED1	- 5mm red LED
LED2	- 5mm green LED
T1-T3	- BC550 npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 4.7-kilo-ohm
R2	- 10-kilo-ohm
R3	- 680-kilo-ohm
R4, R6	- 20-ohm
R5	- 1.8-kilo-ohm
R7	- 330-kilo-ohm
R8	- 47-kilo-ohm
R9, R10	- 620-ohm
R11	- 100-kilo-ohm
R12, R13	- 300-ohm
VR1	- 22-kilo-ohm potentiometer

Capacitors:

C1, C3, C4, C10	- 10 μ F, 16V electrolytic
C2	- 100 μ F, 16V electrolytic
C5	- 22 μ F, 16V electrolytic
C6	- 100pF ceramic disk
C7	- 220 μ F, 16V electrolytic
C8	- 0.33 μ F ceramic disk
C9	- 1nF ceramic disk

Miscellaneous:

MIC1	- Electret microphone
CON1	- 2-pin connector
BATT.1	- 1.5V $\times$ 4 cells

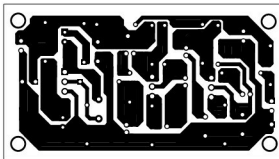


Fig. 2: Actual-size, single-side PCB for noise-level indicator

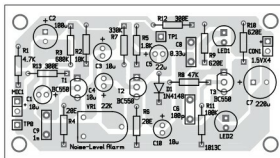


Fig. 3: Component layout for the PCB

a suitable case. Fix LED1, LED2 and potentiometer VR1 on the front panel.

After connecting the 6V power supply to the circuit, set the desired

threshold of sound and adjust VR1 to the point where LED1 starts glowing. For that, switch on radio or TV set and set its volume to a level where you want the warning to start. Now adjust potentiometer VR1 to the point where LED1 starts glowing.

To test the circuit for proper functioning, check correct input supply at TP1 with respect to TP0. LED2 also indicates the same. LED1 glows when the sound level is above threshold, which can be simulated with a radio or music system. ●

The author was a researcher and assistant professor in Technical University of Sofia (Bulgaria), and a lecturer and expert-lecturer in Kingdom of Morocco. Now he is working in the private sector in Bulgaria

Ritchie Street Has Grown Bigger!



Check Out India's leading market-places for trade & commerce in electronics

ele **B2B**.com & **ElectronicsForYou**
YELLOW PAGES

